

BOOK I — FOUNDATIONS: WHAT GRUT IS

WORKING DRAFT — part of the GRUT working corpus; statuses per `books/CORPUS_CHARTER.md` ; subject to chapter-by-chapter audit; nothing here banks.

GRUT Books — Working Edition v1.0 · D. Ryan Grover · 2026-09-07 · release `zenodo-v1.0` · Markdown is the authoritative source; the PDF is generated from it. Drafted from the frozen record by the RAI builder (Claude, Anthropic) under owner direction — this edition is a publishing pass only: formatting, navigation, and metadata; no claim strengthened or weakened for publication.

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Book I of X. This is the front door of the corpus: what GRUT proposes, what it declares as input, what its status hierarchy is and why the tier-marking discipline — not the slogan — is the program's real commitment, and what GRUT is not. Every substantive claim below carries its status inline; the 24 shared claims of the canonical status table appear with their statuses verbatim. Where the record is silent, this book says so. Nothing here is a new derivation; everything restates the frozen record (`GRUT_MODEL_FRAMEWORK.md` · `GRUT_PROGRAM_FREEZE.md` · `provenance/claims.json`, 74 nodes, sha256 `beaeb84e8a6f8468...`, read-only to this corpus).

1 · What GRUT is, in one page

GRUT — historically expanded "Grand Responsive Universe Theory," a name the record itself notes "overpromises more than the program does" (`GRUT_V1_PLAIN.md`) — is a **constitutive-relational framework for physical theory**. Its proposal is a change of primitive language, not a new fundamental Lagrangian: physical systems are to be described through **constitutive response** — what a system does next depends on what has acted on it, through a retarded kernel carrying its interaction history — and the gravitational vacuum itself is treated as such a responsive medium rather than as an inert stage (`GRUT_MODEL_FRAMEWORK.md` §1).

STATUS: ASSUMPTION (AXIOM) — the responsive-medium stance is "a STANCE, explicitly not derived," priced +1 in the register precisely because founding status does not exempt an input from epistemic pricing (source: provenance/claims.json, rung1_ontology_finite_memory, tier assumed, ledger +1).

Three sentences frame everything that follows.

First, GRUT does not claim its primitives are derived from an assumption-free substrate. It is carried, by owner decision, at the Newtonian standard: *given these declared primitives, do the relationships among them produce correct physics, and eventually a distinguishing prediction?* Primitives are not a defect; **undeclared or re-billed primitives are** (GRUT_PROGRAM_FREEZE.md §2).

Second, the program's deepest commitment is not responsiveness. It is the **discipline about responsiveness**: every claim tiered, sourced, priced, and falsifiable, with the tiering enforced by code rather than prose, and with the program's own failures kept on the record's face rather than edited away. The premise a reader should carry out of this book is exactly that pairing — *a responsive vacuum with finite memory, where the ruthless tier-marking matters more than the slogan* (CHARTER.md §§0–1, 6).

Third, GRUT's derivational ladder is **frozen** (owner decision, 2026-09-06). The framework itself is live as a presentation and comparison object, but no new foundational generation may be created "merely because the previous one failed"; the reopening keys are specific, listed, and "hung where the program cannot reach them by itself" (GRUT_PROGRAM_FREEZE.md §§1, 5).

A reader who wants the single most compressed honest summary of the physics is given, by the freeze itself, one sentence with its grades welded on: *on a de Sitter background, persistence came free; forgetting and direction were always paid for* — the first clause a controlled derivation conditional on the background, the second an induction over the examined record, whose universal form is exactly what a one-sided apparatus is least able to check (GRUT_PROGRAM_FREEZE.md §7).

2 · The constitutive-relational worldview

Physics as ordinarily organized is **entity-first**: things exist, and externally specified laws say how they move. GRUT proposes to organize the same physics **response-first**: the primitive is the constitutive relation — a system defined against what it responds to, its present behavior conditioned on its history through a retarded, causal kernel. On this reading, dissipation, noise, decoherence, relaxation, and gravitational response are not separate formalisms bolted together but **sector instances of one architecture**: one influence-functional form, one fluctuation–dissipation lock, one passivity structure, one kernel bookkeeping (GRUT_MODEL_FRAMEWORK.md §§1, 6).

Standard-physics background, distinguished as such: the mathematical carrier of this worldview — the Schwinger–Keldysh (in-in) influence functional with a retarded kernel K_R and a noise kernel N — is standard open-quantum-system machinery (Feynman–Vernon, Caldeira–Leggett, nonequilibrium Keldysh field theory). GRUT's own record is emphatic that adopting this carrier earns the framework nothing by itself:

The influence-functional / Feynman–Vernon form.

STATUS: RECOVERED (generic; u1: the form confers no GRUT-specific content) — any local, causal open quantum system yields this form under coarse-graining; GRUT adopts a universal IR language, it does not derive one (sources: `provenance/claims.json`, `u1_form_universality`; `books/CORPUS_CHARTER.md` canonical table item 1).

So where does GRUT's content live? The record's own answer: **entirely in the declared choices of the primitives table (§5 below) and in whatever consequences those choices earn** (`GRUT_MODEL_FRAMEWORK.md` §3). That is the load-bearing honesty of the whole program, and it is why the primitives table, not any equation, is the backbone of this book.

The worldview's central physical bet, stated at the strength the record supports:

The gravitational vacuum is a responsive medium with finite memory.

STATUS: ASSUMPTION, with REVERSED history on its face (the in-house "no memory time" computation was reversed; exact dS free-field theory forces infinite scale-free memory) — the historically central commitment, retained as postulate only; its strongest form (single pole) carries no surviving discriminating consequence (sources: `books/CORPUS_CHARTER.md` canonical table item 18; `GRUT_MODEL_FRAMEWORK.md` §2, §7).

Note what that status block does: it keeps the founding bet **and** its reversal in the same breath. That is deliberate, and it is the pattern for the entire corpus. The premise is not "responsiveness explains everything." The premise is a specific, priced ontological wager plus the commitment to mark, ruthlessly, what the wager has and has not bought (`GRUT_ToE.md` Vol 0 — model history, ratified as current governance by `CHARTER.md` §1).

3 · Where GRUT currently fails or stops — up front, because it is part of the identity

The framework's authoritative presentation carries its failure list as a numbered section, not an appendix, and this corpus does the same. A reader who skips this section has not read GRUT. Verbatim in substance from `GRUT_MODEL_FRAMEWORK.md` §7, with statuses:

(i) **GRUT does not derive its own primitives.** Every discharge attempt on the record — ROOT-1, the structural search, the resurrection campaign, the "Final Boss" audit — ended in *relocation* of the unexplained input, never its elimination. The surviving irreducible residue is one relative datum: the half-line/KMS alignment.

STATUS: ASSUMPTION (one relative datum: the half-line/KMS alignment; five dressings audited, every closure consumed it) — canonical table item 17 (temporal orientation / Past Hypothesis); sources: `GRUT_PROGRAM_FREEZE.md` §3 (AXIOMS), `RAI_FINAL_BOSS.md`.

(ii) **No novel experimentally confirmed prediction exists.**

STATUS: EMPTY (nothing has earned entry; Book IX governs entry) — the PREDICTED set, canonical table item 21. The emptiness has a *derived* structural reason, not an excuse: the framework asserts a class, a class has no scale, and the admissible cone pins no ratio — "every route from this framework to a number runs outside it" (source: GRUT_PROGRAM_FREEZE.md §3, PREDICTED).

The standing empirical audit is consistent with this and blunter: at interpretation level, GRUT is **signature-null** — it makes no clean, parameter-free, falsifiable prediction that standard physics does not already make.

STATUS: CLOSED (gate outcome) — audit-critic verdict EMPTY across every candidate differentiator channel (GW dissipation, $w(z)$, tabletop decoherence, linear cosmology, founding screens), each closed by suppression, laundering-refusal, or no-go export; one declared soft spot (BH ringdown/QNM, invisible-by-inheritance, not a dedicated calc) (source: SIGNATURE_AUDIT.md).

(iii) **The strongest historical claim was reversed.** The finite-memory kernel's sole in-house quantitative support fell to a real bug validated only at its blind point; corrected, the computed object tends to a constant or grows, and single-pole retains no surviving observable consequence distinguishing it from the passive class (status carried above, §2; see also canonical item 4 in §5's table notes).

(iv) **Several apparent structural results were supplied structure**, identified as such by the program's own audits: the u4 "dissolution" (relocation, not derivation), G-STRONG, the sector-selection and CPR routes, the P1 emergence framing — the last closed by a genuine theorem that cuts *against* the program's earlier reading:

STATUS: DERIVED (RRT arm: intervention response reducible for all linear dynamics; escaped only by nonlinearity) — the linear-universe response no-go, canonical table item 20; the record's cleanest theorem-grade product, and it is deflationary (sources: GRUT_PROGRAM_FREEZE.md §3 DERIVED; §4).

(v) **The audit apparatus itself was caught three times** — an asymmetric error budget, story-pinned gates, a frame-entailed null — and the record's negative results carry the corresponding caps (GRUT_MODEL_FRAMEWORK.md §7; GRUT_PROGRAM_FREEZE.md §1). The program treats "the instrument manufactured part of its own trail" as a finding about itself, on the record, with the contaminated instruments preserved unpatched and their fan-out flagged (GRUT_PROGRAM_FREEZE.md GOVERNANCE EXIT).

These five are not concessions extracted by critics. They are the framework's own §7, and under this corpus's charter they travel with every presentation of GRUT.

4 • What GRUT is NOT

Not a Theory of Everything. The ToE framing is retired, and has been since the clean rebuild's first page: "GRUT is **not a Theory of Everything**, and the program has **retired the 'derive everything' frame**. The filename GRUT_ToE.md is legacy (kept for link stability)" (GRUT_ToE.md, front matter — **model history**; the retirement is current record, restated by CHARTER.md §0: "This is not a theory of everything, and any draft that starts sounding like one has drifted"). A prior book-length v4.0 ToE

deposit exists on Zenodo; the repo's own public documentation states plainly that it "is not the framework described here, and it is substantially wrong" (docs/WHERE_IT_STOPS.md — model history). What replaced the ToE frame, historically, was a closure/universality question — *given responsiveness + locality + causality + open-system dynamics, what is forced and what is imported?* (GRUT_ToE.md — model history) — and what that frame matured into, as of the freeze, is the constitutive-relational framework of this corpus (GRUT_PROGRAM_FREEZE.md §2).

Not a predictive theory, as of this record. The PREDICTED set is **EMPTY (nothing has earned entry; Book IX governs entry)** (canonical item 21, status carried in §3). The word "prediction" is reserved, corpus-wide, for gate-earned results only (books/CORPUS_CHARTER.md).

Not a derivation of standard physics. Where GRUT reproduces GR, quantum mechanics, or standard open-system results, the record marks the reproduction as recovery, with the honesty note attached:

STATUS: RECOVERED (largely by identity; KERNEL-STANDARD, scoped) — GR at zero-memory collapse, canonical table item 19; recovery-by-identity is compatibility, not correspondence evidence, because the executed machinery *is* standard machinery on the declared inputs (sources: GRUT_PROGRAM_FREEZE.md §3 RECOVERED; GRUT_MODEL_FRAMEWORK.md §4).

Not self-certifying about its own method. The discipline apparatus — register, ledger, gates, adversarial screens — is the program's candidate contribution, and it is itself held to the tier discipline:

STATUS: UNRESOLVED — the method's novelty is graded in-register as a weakly novel synthesis of established methodology (every component prior art), with the "biggest contribution" over-claim retracted and independent external validation owed; a claim cannot be banked on self-verification (source: provenance/claims.json , method_novelty , tier to-derive).

Not an ontology to defend. The governing V2 charter is explicit: every branch is a constrained classification problem with first-class failure states — "never an ontology to defend" (CHARTER.md §8). Speculative furniture — the responsive-vacuum picture as organizing intuition, self-recording/universal-refresh, black-hole saturation, "1Space" — is permitted to exist and **barred from load-bearing**; 1Space in particular is UNDEFINED on the record, all seven candidate definitions having failed non-circularity (GRUT_PROGRAM_FREEZE.md §3, SPECULATIVE).

Not a ladder that keeps climbing. The stopping rule, owner's words, committed governance: "No new RRT generation, model class, or foundational rung will be created merely because the previous one failed" — a new generation requires an independently motivated principle naming a specific phenomenon the existing framework cannot represent (GRUT_PROGRAM_FREEZE.md §1).

5 · The explicit primitives — what GRUT declares as input

This table is the backbone of the framework and of this book: the complete declared-input sheet of GRUT_MODEL_FRAMEWORK.md §2 ("nothing hides in the mathematics"), reproduced faithfully with the canonical statuses attached. The eight-status vocabulary is defined in §7 below; ASSUMPTION rows carry their register kind (AXIOM / EMPIRICAL-INPUT / STRUCTURAL-SELECTION) because the kinds are different epistemic objects.

#	element	status	why selected	alternatives considered / fate
P1	responsiveness (constitutive relation as primitive)	ASSUMPTION (AXIOM)	the central ontological bet; a framework must bet something	entity-first ontology is the standard alternative; GRUT's claim is that the response language organizes better
P2	relational description (system defined against what it responds to)	ASSUMPTION (AXIOM)	inseparable from P1	the audits showed the <i>specific</i> decomposition is a further choice (P3)
P3	a system/bath-type decomposition	ASSUMPTION (AXIOM (form) + STRUCTURAL- SELECTION (instance))	required to write any influence functional	the canonical tensor split is <i>impossible</i> in the relevant algebra type (III ₁); what survives translation is a one- sided inclusion; the contract's operative partition (external- leg vs internal-line) was historically undeclared — now declared
P4	memory / finite relaxation	ASSUMPTION, with REVERSED history on its face (the in-house "no memory time" computation was reversed; exact dS free- field theory forces infinite scale-free memory)	the historically central bet	retained as postulate only; its strongest form (single pole) carries no surviving discriminating consequence
P5	retardation / causality of K_R	ASSUMPTION (EMPIRICAL-INPUT + STRUCTURAL)	observed time- asymmetry; upper- half-plane analyticity	the audits located its origin in a past-boundary condition — an input in every formulation examined
P6	Past Hypothesis (direction of relaxation)	ASSUMPTION (one relative datum: the half- line/KMS alignment; five dressings audited, every closure consumed it)	no examined route derives temporal orientation	five dressings audited; all one input
P7	Born measure	ASSUMPTION (inherited axiom; the improper- mixture objection stands)	decoherence selects a basis, not a probability	inherited, irreducible in- house; flagged in-register as "the never-sorted door"
P8	bath content (massless relativistic modes)	ASSUMPTION (STRUCTURAL- SELECTION)	simplest relativistic choice	classified "standard input" by the distinctiveness ledger; the proviso "no second internal scale" is undischarged

#	element	status	why selected	alternatives considered / fate
P9	TT projector (pure spin-2 response)	ASSUMPTION (STRUCTURAL-SELECTION — CHOSEN, unanimous five-angle interrogation; not forced)	diffeo invariance stops one condition short of forcing it	the {shear, bulk} interior remains the unexplored empirically viable sector
P10	$x = 0$ scalar-to-tensor choice	ASSUMPTION (STRUCTURAL-SELECTION; $x_{\text{no_pin}}$: the action carries a family, pins nothing)	the working point of the cosmology exports	the admissible set is an amplitude-homogeneous cone — no ratio pin exists to force any x
P11	$c_0 = \alpha$ DC normalization	ASSUMPTION (STRUCTURAL-SELECTION)	adopted phenomenological anchor	the conformal anomaly does not derive it (rung9b settled-negative on three named obstructions)
P12	state (BD-analogue), scheme, order of limits, gauge — the D1–D5 declaration sheet	ASSUMPTION (STRUCTURAL-SELECTIONS)	each declared and priced	the crossed-product result shows state-selection dissolves at that level (Connes cocycle)
P13	measured constants (H_0 , M_P , T_{dS} given background, ...)	ASSUMPTION (EMPIRICAL-INPUTS)	nature's numbers	see the T_{dS} note below — the register's only credit

(Row sources: `GRUT_MODEL_FRAMEWORK.md` §2 throughout; register nodes `rung1_ontology_finite_memory` (P1/P4), `u3_split_origin` (P3), `past_hypothesis / arrow_of_time` (P5/P6), `born_rule` (P7), `rung3_single_pole` boundary condition (P8), `p_tt_ansatz` (P9), `x_no_pin_theorem` (P10), `rung9b_bridge` (P11).)

Five rows deserve prose, because they are where readers most often mis-take an input for a result or vice versa.

P3, the split (the deepest assumption). Any influence functional presupposes a system/bath split; the split itself is nowhere derived, and the register fences the open question against pre-answering: *why is there a split at all* is `u3_split_origin`, held open with a machine-checkable fence — "a screen returning 'looks emergent' does NOT bank 'emergent'; only an exhibited derivation graduates this, in either direction" (`provenance/claims.json`, `u3_split_origin`, tier `to-derive`, fenced). The audits also sharpened what kind of thing the split can even be in the relevant operator algebra: not a tensor factorization (type III₁ forbids it) but a one-sided inclusion — and they caught the contract's operative partition as historically undeclared, now declared (`GRUT_MODEL_FRAMEWORK.md` §2; `GRUT_PROGRAM_FREEZE.md` §3 AXIOMS).

P4 and its strongest form. The single-pole / finite-memory kernel — historically billed as GRUT's one novel structural commitment — is today graded:

STATUS: UNRESOLVED (anchor-class, derived-pending; pole-vs-cut open; the Tier-4 computation found a CUT, not a pole, at flat scope) — canonical table item 4; the register's own tier note warns equally against over-claiming it and over-demoting it (sources: `provenance/claims.json`, `rung3_single_pole`; `books/CORPUS_CHARTER.md` item 4).

P6, the direction of time. GRUT's honest product here is a *decomposition*, not a derivation: the in-in machinery intrinsically supplies the **existence and magnitude** of dissipation (retarded analytic self-energy, positive spectral measure), while the **direction** is imported from the bath's low-entropy past — like every program since Boltzmann. The register grades the whole as `assumed`, +1, and names the contribution as methodological: locating exactly where and in what form the arrow is assumed (`provenance/claims.json`, `arrow_of_time`, `past_hypothesis`).

P9, the pure-TT projector. This is the register's single most interrogated input and the model case for the STRUCTURAL-SELECTION category: a five-angle, adversarially verified, exact-rational-arithmetic interrogation asked whether the declared structure *forces* the projector, and answered — unanimously — **CHOSEN, not forced**. The linear-cosmology statement rests on it; that is why the recovery of Λ CDM is stated as conditional (`provenance/claims.json`, `p_tt_ansatz`; `BRIEF_p_tt_interrogation.md`).

P13, and the one credit. Among the empirical inputs one item is *not* free:

STATUS: DERIVED (within declarations: forced by Hadamard/KMS on the declared background) — $T_{dS} = H/2\pi$, canonical table item 3: given the declared background, the temperature carries zero freedom; the register calls this its only credit of the kind (sources: `books/CORPUS_CHARTER.md` item 3; `provenance/claims.json`, `rung2_kms_gate` ledger note).

Finally, one constraint that binds the primitives together rather than adding to them:

The KMS/FDT lock.

STATUS: ASSUMPTION (borrowed standard identity, enforced as a hard admission gate; rung2) — in equilibrium the noise kernel is locked to the dissipation by detailed balance; the lock is standard physics (Callen–Welton, Kubo), and GRUT's move is to enforce it as an executable admission gate (`gate/kms.py`) that bars any candidate kernel failing it (sources: `books/CORPUS_CHARTER.md` item 2; `provenance/claims.json`, `rung2_kms_gate`).

The reader should now be able to answer the front-door question precisely: **GRUT is the conjunction of rows P1–P13, the KMS gate, and the standard influence-functional carrier — plus whatever consequences that conjunction has demonstrably earned.** The earned list is short and lives in Books II–VI; its headline entries as of the freeze are the flat-scope one-loop TT kernel with $H^1 = 0$ and the even-degree ladder class, the $s = 5$ spectral law at flat scope (rejecting the framework's own registered $s = 3$), the exact- dS constant tail $H^2/4\pi$, the passivity/cone theorems, the linear-universe response no-go, and $T_{dS} = H/2\pi$ (`GRUT_PROGRAM_FREEZE.md` §3, DERIVED). One of these carries a status worth stating here because it disciplines how the reader should weight even the genuine mathematics:

STATUS: DERIVED (the one genuinely new structural identity; carries no confirmatory weight for GRUT) — the $H^1 = 0$ four-channel cancellation and its even-degree ladder class (Theorem II), canonical table item 24: new mathematics, honestly decoupled from the framework's truth (source: books/CORPUS_CHARTER.md item 24).

6 · The status hierarchy — the eight-status ledger, and why it is the real commitment

The vocabulary. As of the freeze, every claim in the program is carried in an eight-status vocabulary, replacing the earlier five-tier register vocabulary and its "shame-tiering" connotations (GRUT_PROGRAM_FREEZE.md §2):

AXIOM · EMPIRICAL-INPUT · STRUCTURAL-SELECTION · DERIVED · RECOVERED · PREDICTED ·
UNRESOLVED · SPECULATIVE

The corpus's display labels (this book's bold status tags) map onto that ledger per books/CORPUS_CHARTER.md: ASSUMPTION covers the three input kinds and always says which; HYPOTHESIS displays SPECULATIVE; two record-outcome labels (CLOSED for computed refusals, REVERSED for in-house results later overturned) and one absence label (UNMAPPED) complete the set. Two rules give the vocabulary its teeth. **Scope qualifiers travel with statuses** — "DERIVED (flat contract scope, $\omega \gg H$)" is a different claim from "DERIVED," and dropping the parenthesis is how honest results become dishonest ones. And **PREDICTED is gate-earned only**; it is currently empty, and no display label may drift a HYPOTHESIS toward it.

The machinery. The vocabulary would be decoration if it lived in prose. It does not. The program's ground truth is a machine-checked register — provenance/claims.json, 74 nodes — in which every claim carries a tier, sources that must resolve, a named *overturning computation* (what would falsify it), and a signed ledger_delta pricing the assumptions it adds or removes; the net is a blind sum printed by provenance/validate.py, deliberately never hand-quoted in standing documents because hand-typed copies go stale (GLOSSARY.md, translation layer). An anti-laundering gate blocks any net-positive claim sold as a derivation (provenance/auditor.py); a live bank-time gate diffs every proposed register edit against the last accepted baseline and surfaces substantive changes for adversarial screening before a human accepts them (provenance/bankgate.py); and disposition markers must live in the machine-scanned sub_status field, not in prose, because a disposition the resident cannot see is a disposition that can silently soften (CHARTER.md §7). Pre-registration is house rule: expectations are written and hashed before an analysis runs, results live in a separate citing file, and the termination condition itself was pre-registered outcomes-first and owner-signed (CHARTER.md §1.6).

The calibration that matters. The record states it about itself, and this corpus repeats it wherever the machinery is described: this apparatus verifies **discipline, not truth**. A wrong-but-well-provenanced claim passes; the blind ledger sum cannot catch a physics error or a double count. The claim made for the register is that its contents carry *provenance*, never asserted *correctness* (GRUT_ToE.md Vol 0 — model history; RESULTS_auditor.md).

Why this is the real commitment. The record's answer is empirical, about itself. The discipline is what caught the founding bet's collapse (P4's reversal was found by the program's own instruments and kept on its face); it is what demoted the program's favorite result rather than graduating it (`run3` held at anchor-class against pressure in both directions); it is what turned up the $\mu = 4/3$ self-exclusion —

STATUS: CLOSED (self-exclusion: separate-universe consistency + low- ℓ ISW) — the trace-only endpoint GRUT's own conformal coefficient naively suggests is excluded, a genuine self-exclusion the framework earned (source: `books/CORPUS_CHARTER.md` item 14)

— and it is what the one blind external-style adjudication on the record identified as the program's actual product: "the transferable asset is the method, not the physics" (`GRUT_PROGRAM_FREEZE.md` §7). The same discipline polices its own language: the register's historical uses of the word "specialist" were audited, found to denote in-house AI-relayed passes in the dangerous cases, and a standing rule now bars the unqualified word from any public document (`GLOSSARY.md`). A program that does that to its own vocabulary is making a specific claim about what it values; that claim — not the responsive vacuum — is what the record most strongly supports.

The five-tier build vocabulary (shown / derived / derived-pending / assumed / to-derive) remains the register's native grading and appears verbatim wherever this corpus quotes register nodes; the charter mapping in `books/CORPUS_CHARTER.md` is the crosswalk, and no node-by-node re-grading is performed by this corpus.

7 · Scope: what GRUT is attempting to describe — and the absence map

In scope, with banked content (Books II–VI, VIII). The gravitational vacuum's linear-response structure (the retarded TT kernel at its declared contract scope, $\omega \gg H$); equilibrium noise via the KMS/FDT lock; decoherence and open-system reduction on declared inputs; the arrow-of-time decomposition; background cosmology at the chosen point —

STATUS: DERIVED (within the choices $x = 0$ / pure-TT: the sourced cosmology statement) — $w = -1$ flat: linear cosmology reproduces Λ CDM at the declared working point; the conditionality on P9/P10 is part of the claim (source: `books/CORPUS_CHARTER.md` item 13, first clause)

— and one proposed phenomenological signature, graded as such:

STATUS: HYPOTHESIS (proposed phenomenological relation; shape-only; magnitude verdict quiet-or-faint) — the USL gravitational-decoherence shape signature, canonical table item 12; a proposed relation, not a derived prediction, not a validated effect (source: `books/CORPUS_CHARTER.md` item 12).

In scope but currently unaskable. The deep-infrared regime $\omega \lesssim H$ is UNRESOLVED in a specific, honest sense: **UNASKABLE at current declarations, on four obstructions — not false** (`GRUT_PROGRAM_FREEZE.md` §3, UNRESOLVED). The kernel-vs-dressed-object identity of the registered Tier-4 result is an open owner fork. `u3` (why a split at all) and `u4` (why a constitutive form at all) are open and fenced in both directions.

Borrowed and marked. Where GRUT touches deep structure it does not own, the record marks the borrowing:

STATUS: RECOVERED (borrowed; audited verdict B-INPUT-RELOCATION: the unexplained input moves, it is not discharged) — the type $III_1 \rightarrow II_1$ crossed-product construction (CLPW), canonical table item 23 (source: books/CORPUS_CHARTER.md item 23).

Out of scope — the absence map. The record contains **no GRUT account at all** of the following, and this corpus will not manufacture one:

STATUS: UNMAPPED — flavor, strong-CP, neutrino masses, dark matter, baryogenesis (canonical table item 22; source: books/CORPUS_CHARTER.md ; the standing memory record of the register likewise carries "flavor & strong-CP unmapped" as a prerequisite gap).

Also absent from the record, stated here so the absence is on this book's face: any GRUT account of particle-physics dynamics beyond the interface statuses of Book VII; any black-hole ringdown/QNM calculation (the signature audit's one declared soft spot, invisible-by-inheritance only); any definition of "1Space" that survives non-circularity; and any derivation of the Born measure, the split, or temporal orientation (§5). Aspirational extensions that appeared in the historical book — complexity, life, consciousness — were open research program, never established content, even there (GRUT_ToE.md — model history), and are nowhere in the frozen record as physics.

8 · What would change the program's mind

Foundations end with falsifiability, because the framework's §8 does. GRUT is **strengthened** by: the interacting graviton zero-mode computation (O2) returning *protected*; a derivation of any §5 selection from an independently motivated principle; a quantitative sector consequence not encoded in the inputs; the SLOT test resolving the clock-slot as not a second irreducible input. GRUT is **weakened or falsified** by: an observed $w = -1$ crossing (which kills the entire passive relaxational class at a stroke, GRUT included); O2 returning *lifted*; a self-closing description of the observed time-asymmetry with zero undischarged inputs; the USL shape signature excluded in its distinguishable regime; or the RESIDUE test deriving the half-line/KMS alignment from unoriented hypotheses (GRUT_MODEL_FRAMEWORK.md §8; reopening conditions, GRUT_PROGRAM_FREEZE.md §5). The freeze's closing sentence belongs in the foundations and so closes this book's presentation of them: *nature and mathematics hold every one of these votes; the framework holds none of them.*

The near-term program that operates under these foundations — consolidation, targeted verification, sector-by-sector comparison, the prediction hunt with its own stop rule ("if no discriminator survives — say so, in the model document, and stop") — is recorded in GRUT_NEXT_STEPS.md and taken up in Books IX and X.

Sources drawn from

- `books/CORPUS_CHARTER.md` — vocabulary, canonical status table, formatting (binding)
- `GRUT_MODEL_FRAMEWORK.md` — §§1–8; the §2 primitives table is this book's backbone
- `GRUT_PROGRAM_FREEZE.md` — stopping rule, reclassification, eight-status ledger, §3 ledger, §§5–7
- `CHARTER.md` — register governance, the five disciplines, §7 resident rule, §8 u0 charter
- `GRUT_NEXT_STEPS.md` — the practical program and its stop rule
- `GRUT_MINIMUM_CORE.md` — the minimum-core framing and its exclusions
- `SIGNATURE_AUDIT.md` — the signature-null verdict, channel table, the QNM soft spot
- `GRUT_ToE.md` — front matter and Vol 0–1 (**model history**: ToE retirement, closure frame, discipline ethic)
- `GRUT_V1_PLAIN.md` — the name expansion and its caveat (line 11)
- `docs/WHERE_IT_STOPS.md` — the v4.0 Zenodo deposit disclaimer (model history)
- `GLOSSARY.md` — translation layer; the "specialist" vocabulary audit and standing rule
- `provenance/claims.json` — nodes read directly: `rung1_ontology_finite_memory`, `rung2_kms_gate`, `rung3_single_pole`, `arrow_of_time`, `past_hypothesis`, `born_rule`, `p_tt_ansatz`, `x_no_pin_theorem`, `u1_form_universality`, `u3_split_origin`, `u4_constitutive_origin`, `method_novelty`, `background_time_translation_flow`, `response_lorentz_covariance`, `kr_contract_retarded_tier4`

Gaps in this book

1. **The primitives table's "why selected" column is reproduced from `GRUT_MODEL_FRAMEWORK.md` §2 prose**; the underlying interrogation artifacts (e.g. `BRIEF_p_tt_interrogation.md`, the D1–D5 declaration sheet's own file, the crossed-product audit) are cited but not re-narrated here — Book VIII carries the algebraic material.
2. **No node-by-node crosswalk** between the register's native five-tier vocabulary and the eight-status ledger is attempted; only the charter's mapping is used. Where an individual node's crosswalk could be contested (e.g. `to-derive` rendered as UNRESOLVED for `method_novelty`), this book's rendering is a display choice, not a re-grading.
3. **Model history before the clean rebuild** (the v2/v3/v4 lineages, the Zenodo v4.0 deposit's content) is not reconstructed; `GRUT_ToE.md` and `docs/WHERE_IT_STOPS.md` were consulted for framing history only, and the record here is silent on the interior of those lineages by design ("earlier results have no standing until re-derived").
4. **The RRT-0 / RAI instrument arc** (the five consecutive no's, the dialectic chamber, the Final Boss residue) is cited for its verdicts only; the RRT-0/RRT-1 narration lives in Book III §8 and the instrument-arc consolidation in Book X §6 (pointer corrected at corpus audit).
5. **The {shear, bulk} interior sector** (the empirically viable family P9 forecloses) is flagged but not developed; its computed window lives in `SIGNATURE_AUDIT.md` and `zeta_interior_family`, and belongs to Book VI.
6. **The record is silent** — and so is this book — on: flavor, strong-CP, neutrino masses, dark matter, baryogenesis (UNMAPPED); any QNM/ringdown calculation; any non-circular definition of 1Space; any derivation of the Born measure, the system/bath split, or temporal orientation.

7. **The method-novelty question cannot be settled from inside:** the register itself rules that self-validation is insufficient, so this book reports the method's status (UNRESOLVED, external validation owed) and cannot do more.
8. **Current ledger net figures are deliberately not quoted** (they go stale in prose, per the program's own recorded defect); the live number is `provenance/validate.py`'s output.